The 24-Meet of Lo Shu and Dürer: Bounded Magic Spectra, Sagrada Rays, and F_2^4 Transport

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Abstract

We isolate a bounded endpoint shared by two different magic-square mechanisms. On the 3×3 side, repeated-entry magic squares with entries in $[1, 9]$ have an upper non-degenerate Lo Shu fiber at sum 24 and a degenerate all-9 fiber at sum 27. On the 4×4 side, a one-incidence Sagrada mask on the Dürer-complement square defines a bounded ray $D(t) = D - tM$ whose terminal point occurs at $t = 10$, again at sum 24. The weak meet is $\{24, 27\}$; after imposing non-degeneration on the Lo Shu side and terminality on the Dürer/Sagrada side, the unique strong meet is $\{24\}$.

The endpoint also carries finite structure on the Dürer side. The target permutation diagonals drop from a dihedral subgroup $D_4 \leq S_4$ to the Klein four subgroup V_4, and the Dürer value-minus-one cell model gives an F_2^4 affine transport layer: 52 affine source quaternes reduce to 36 affine terminal quaternes. A broader order-four audit shows that terminal 24 is not unique: it occurs in 236 square-mask pairs. Inside that landscape, the most stable finite class detected by the atlas is a 144-pair exact V_4 subclass characterized by global F_2^4-affineness; selected-affine near misses are separated inside the main 176-pair class by a quotient-shadow obstruction. Thus 24 is not a universal invariant of magic squares. It is a structured endpoint in a specific bounded meet, supported by exact finite certificates.

Keywords. Magic squares, Lo Shu, Dürer square, Sagrada Familia square, bounded lattice points, permutation diagonals, Klein four group, affine geometry over F_2, finite certificates.

MSC 2020. 05B15 (primary), 05A15, 05E18, 11P21, 52B20, 20B25.

1 Introduction

The classical Lo Shu square and Dürer's order-four magic square belong to different corners of the magic-square literature. Dürer's square has a rich record of symmetries, quaterne patterns, and Sagrada Familia variants [Wei26, Mar01, TT13, Sud13]. Order-four magic squares also have a classical finite classification: up to the standard equivalences there are 880 normal representatives [OB82]. The present paper adds no universal numerological slogan to this literature. It puts one observation into a bounded, certificate-controlled category.

The observation is that the value 24 occurs as the endpoint of two unrelated-looking bounded mechanisms:

- a non-degenerate upper fiber of repeated-entry 3×3 magic squares with values in $[1, 9]$;
- the terminal point of a one-incidence Sagrada perturbation ray inside a 4×4 Dürer-complement square with values in $[1, 16]$.

The first mechanism is a lattice diamond. The second is a bounded ray. Their weak numerical intersection contains 24 and 27, but 27 fails both endpoint tests: it is the degenerate all-9 Lo Shu fiber and an interior point of the Dürer/Sagrada ray. The strong meet is therefore 24.

The manuscript has two layers of results.

(a) The central bounded-meet theorem is a short mathematical consequence of the two certified spectra.

(b) The Dürer-side structure is an exact finite replay: quaterne transport, the $D_4 \to V_4$ permutation-diagonal drop, the $\mathbb{F}_2^4$ affine transport layer, the bounded-polytope guardrail, and the order-four endpoint atlas.

We use an explicit claim-level convention: replay-backed statements are labelled as *Certified Propositions*. These are not external theorems; they are finite exact replays or finite enumerations whose scope is explicitly bounded by repository artifacts, tests, and a claim crosswalk. The controlling ledger for this manuscript is

`docs/CLAIM_CROSSWALK.md`,

and the main certificate pack is

`results/magic24_certificate_pack.json`.

What is not claimed. The guardrails are part of the result. The paper does not claim that 24 is a universal invariant of magic squares, that the terminal square $D(10)$ is more perfect than Dürer's source square, or that $D(10)$ is a vertex of the bounded order-four magic polytope. It also does not identify the terminal V_4 diagonal subgroup with a type-A poset cone, and it does not claim that the $\mathbb{F}_2^4$ layer isolates only the historical Dürer/Sagrada cell in the full order-four atlas. These stronger statements are listed as blocked claims in the reproducibility package.

Organization. Sections 2–5 prove the strong bounded meet. Sections 6–8 give the Dürer-side enrichment: quaterne transport, permutation diagonals, and the $\mathbb{F}_2^4$ tesseract layer. Section 9 records the bounded-polytope guardrail, and Section 10 summarizes the order-four extension. The appendices collect the endpoint atlas, exact canonical V_4 subclass, secondary audits, and open problems.

2 Bounded Magic Squares and One-Incidence Rays

Definition 2.1 (Bounded repeated-entry magic squares)**.** For positive integers n, m, S, let $\mathcal{B}(n, m, S)$ be the set of integer $n \times n$ arrays $A = (a_{ij})$ with entries in $[1, m]$ such that all rows, all columns, and the two main diagonals sum to S. Repeated entries are allowed.

The repeated-entry convention is essential. It lets the Lo Shu side form a short bounded spectrum and lets the Dürer side move along a perturbation ray. Thus the phrase "Lo Shu side" means the affine 3×3 magic-square family normalized around the Lo Shu center, not normal 3×3 squares with distinct symbols $1, \dots, 9$. It is separate from the distinct-entry inside-out enumerations of magic and magilatin labellings [BZ06b, BZ06a, BVH11]. Throughout this paper, "magic" means row sums, column sums, and the two main diagonal sums; panmagic or toroidal diagonal constraints are not part of the ambient category unless explicitly stated.

Definition 2.2 (One-incidence mask)**.** Let Q be an $n \times n$ magic square. A zero-one mask M is one-incidence if every row, every column, and both main diagonals of M have sum 1. Then

$$Q(t) = Q - tM$$

is again magic, with magic sum decreased by t, for every integer t. The bounded terminal time is the largest $t \geq 0$ for which $Q(t)$ remains inside the prescribed entry interval.

For 4×4 squares, one-incidence masks are indexed by the permutations $\sigma \in S_4$ that use one cell in each row and column and also one cell in each main diagonal.

3 The Lo Shu Lattice Diamond

Every 3×3 magic square can be parametrized as

$$\begin{pmatrix} g+a & g-a-b & g+b \\ g-a+b & g & g+a-b \\ g-b & g+a+b & g-a \end{pmatrix}, \qquad S = 3g.$$

Imposing entries in $[1, 9]$ gives a lattice diamond in the (a, b)-plane:

$$|a+b| \le m, \qquad |a-b| \le m, \qquad m = \min(g-1, 9-g).$$

The diamond has

$$2m^2 + 2m + 1$$

integer points.

Certified Proposition 3.1 (Bounded Lo Shu spectrum)**.** *For $S = 3, 6, \ldots, 27$, the sizes of $\mathcal{B}(3, 9, S)$ are*

$$1,\ 5,\ 13,\ 25,\ 41,\ 25,\ 13,\ 5,\ 1.$$

In particular,

$$\mathrm{Spec}^{+}_{3,9}(15) = \{18, 21, 24, 27\}.$$

At $S = 24$ the fiber has 5 points, while at $S = 27$ the fiber is the unique all-9 square.

Certification. This is Claim IDs M24-C01–M24-C03 and M24-C57 in `docs/CLAIM_CROSSWALK.md`. The replay is in the certificate pack and the Lo Shu lattice-polygon note:

- `results/magic24_certificate_pack.json`
- `docs/LO_SHU_LATTICE_POLYGON.md`

The formula above gives the closed-form count. □

S	3	6	9	12	15	18	21	24	27
$\lvert\mathcal{B}(3, 9, S)\rvert$	1	5	13	25	41	25	13	5	1

Table 1: Bounded repeated-entry 3×3 magic-square counts with entries in $[1, 9]$.

At $S = 24$, $g = 8$, $m = 1$, and the five parameter points are

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (-1, 0), \quad (0, -1).$$

At $S = 27$, $g = 9$, $m = 0$, and the diamond collapses to a single point. This collapse is the non-degeneration distinction that will remove 27 from the strong meet.

4 The Dürer/Sagrada Ray

We use the Dürer-complement square

$$D = \begin{pmatrix} 1 & 14 & 15 & 4 \\ 12 & 7 & 6 & 9 \\ 8 & 11 & 10 & 5 \\ 13 & 2 & 3 & 16 \end{pmatrix}, \qquad S_D = 34.$$

$$\begin{pmatrix} 1 & 14 & \boxed{15} & 4 \\ \boxed{12} & 7 & 6 & 9 \\ 8 & \boxed{11} & 10 & 5 \\ 13 & 2 & 3 & \boxed{16} \end{pmatrix}$$

Figure 1: The Sagrada mask 2013 on the Dürer-complement square.

This is the complement orientation of the classical Dürer square, obtained by $x \mapsto 17 - x$, chosen because it matches the Sagrada mask convention used here. Let M be the one-incidence mask indexed by 2013, selecting the four cells

$$(0,2), \quad (1,0), \quad (2,1), \quad (3,3).$$

These cells contain $15, 12, 11, 16$.

Certified Proposition 4.1 (Sagrada terminality)**.** *The ray $D(t) = D - tM$ remains inside $[1, 16]$ exactly for*

$$0 \le t \le 10.$$

Its terminal magic sum is $34 - 10 = 24$. Among the eight one-incidence Dürer masks, 2013 is the unique mask with terminal sum 24. Equivalently, it is the unique maximin mask: it uniquely maximizes the smallest selected entry, hence uniquely maximizes the allowed downward time $t_{\max}$.

Certification. For selected values $v_1, \ldots, v_4$, the bound $D - tM \in [1, 16]$ gives

$$t_{\max} = \min_i (v_i - 1).$$

The selected values for 2013 are $15, 12, 11, 16$, so $t_{\max} = \min(14, 11, 10, 15) = 10$. The eight-mask replay in Table 2 shows that no other admissible mask has selected minimum 11, so this is also the unique maximin mask. The full replay is Claim IDs M24-C04, M24-C05, M24-C51, and M24-C64, certified in `results/magic24_certificate_pack.json` and `results/f2_tesseract_analysis.json`. □

Mask	Selected values	$t_{\max}$	Terminal sum
0231	$1, 6, 5, 2$	0	34
0312	$1, 9, 11, 3$	0	34
1203	$14, 6, 8, 16$	5	29
1320	$14, 9, 10, 13$	8	26
2013	$15, 12, 11, 16$	10	24
2130	$15, 7, 5, 13$	4	30
3021	$4, 12, 10, 2$	1	33
3102	$4, 7, 8, 3$	2	32

Table 2: The eight admissible one-incidence masks for the Dürer-complement orientation used in this paper.

The closed bounded ray spectrum is

$$\mathrm{Ray}^{-}_{D,M} = \{24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34\}.$$

For the meet calculation, we use the strict downward part below the source sum 34, namely $\{24, \ldots, 33\}$, because the meet compares Lo Shu values above the base sum 15 with nontrivial downward perturbations below the Dürer source.

Lemma 4.2 (Terminal feasibility)**.** *Let A be any normal order-four magic square, with entries $\{1, \ldots, 16\}$, and let N be a one-incidence mask selecting four cells. If m is the smallest selected entry, then the terminal sum of the bounded descent $A - tN$ is $35 - m$. Consequently, among the upper Lo Shu sums*

$$\{18, 21, 24, 27\},$$

only 24 and 27 can occur as terminal sums of such a one-incidence descent.

Proof. The source magic sum of a normal order-four square is 34. A one-incidence mask subtracts t once from each row, column, and main diagonal, so the magic sum becomes $34 - t$. The lower bound 1 allows

$$t_{\max} = m - 1,$$

and therefore the terminal sum is $34 - (m - 1) = 35 - m$.

To match an upper Lo Shu sum S, one must have $m = 35 - S$. For $S = 18$, this gives $m = 17$, impossible in $\{1, \ldots, 16\}$. For $S = 21$, this gives $m = 14$, but four distinct selected entries all at least 14 cannot be chosen from $\{1, \ldots, 16\}$. The remaining cases $S = 24$ and $S = 27$ require $m = 11$ and $m = 8$, respectively, and are not excluded by this terminal feasibility test. □

5 The Strong Meet

Definition 5.1 (Weak and strong meet)**.** The weak meet is the numerical intersection of the upward Lo Shu spectrum above 15 and the strict downward Dürer/Sagrada ray spectrum below 34. The strong meet keeps only values that are non-degenerate on the Lo Shu side and terminal on the Dürer/Sagrada side.

Theorem 5.2 (Strong bounded Lo Shu/Dürer-Sagrada meet)**.** *For the bounded categories above,*

$$\mathrm{Meet}_{\mathrm{weak}} = \{24, 27\}, \qquad \mathrm{Meet}_{\mathrm{strong}} = \{24\}.$$

Proof. By Certified Proposition 3.1,

$$\mathrm{Spec}^{+}_{3,9}(15) = \{18, 21, 24, 27\}.$$

By Certified Proposition 4.1, the strict downward Dürer/Sagrada ray contains

$$\{24, 25, 26, 27, 28, 29, 30, 31, 32, 33\}.$$

The intersection is $\{24, 27\}$. The value 24 is non-degenerate on the Lo Shu side and terminal on the Dürer/Sagrada side. The value 27 is the all-9 Lo Shu fiber and occurs at the non-terminal Dürer ray time $t = 7$. Therefore only 24 survives the strong meet rule. □

Remark 5.3. The value 27 is the control case. It shows why the theorem needs the two boundary conditions rather than a bare intersection of spectra.

$$\begin{array}{rcl} \text{Lo Shu upper spectrum} & : & 18 \quad 21 \quad \boxed{24} \quad 27_{\mathrm{deg}} \\ \text{Dürer/Sagrada downward ray} & : & 33 \quad 32 \quad \cdots \quad 27 \quad \cdots \quad \boxed{24}_{\mathrm{term}} \\ \multicolumn{3}{c}{\mathrm{Meet}_{\mathrm{weak}} = \{24, 27\}, \qquad \mathrm{Meet}_{\mathrm{strong}} = \{24\}.} \end{array}$$

Figure 2: The strong meet keeps the value that is simultaneously non-degenerate on the Lo Shu side and terminal on the Dürer/Sagrada side.

6 Quaterne Transport

For a 4×4 square A, let $H_s(A)$ denote the family of four-cell subsets whose entries sum to s. These subsets are not required to be rows, columns, diagonals, or geometric lines. They are the four-uniform set-system layer of the square: the transport problem asks how these labelled hyperedges move under a bounded mask perturbation. We use "quaterne" for four-cell subsets, following the Dürer-pattern literature. Dürer's 34-sum quaternes are a classical theme in the literature on melancholic magic squares [TT13].

Certified Proposition 6.1 (Dürer/Sagrada quaterne transport)**.** *The Dürer-complement square has*

$$|H_{34}(D)| = 86.$$

Against the Sagrada mask, those 86 *quaternes have incidence distribution*

$$19,\ 50,\ 17$$

for mask incidences $0, 1, 2$*. The terminal square has*

$$|H_{24}(D(10))| = 96,$$

with source decomposition

$$25 + 50 + 21,$$

coming respectively from source sum 24 *and incidence* 0*, source sum* 34 *and incidence* 1*, and source sum* 44 *and incidence* 2*.*

Certification. This is Claim IDs M24-C07–M24-C09 and M24-C117, replayed in

- `results/magic24_certificate_pack.json`
- `results/inside_out_set_system_audit.json`

The transport rule is immediate from $D(10) = D - 10M$: a source quaterne with Sagrada incidence k has terminal sum decreased by $10k$. $\square$

Source family	Mask incidence	Count	Terminal sum at $t = 10$
$H_{34}(D)$	0	19	34
$H_{34}(D)$	1	50	24
$H_{34}(D)$	2	17	14
Source sum 24	0	25	24
Source sum 34	1	50	24
Source sum 44	2	21	24

Table 3: The source $H_{34}(D)$ incidence split and the terminal $H_{24}(D(10))$ decomposition.

Remark 6.2 (Johnson-scheme reading). The quaternes are 4-subsets of the 16-cell set, hence live in the Johnson scheme $J(16, 4)$. For a selected mask M, write

$$J_i(M) = \{Q : \ |Q| = 4,\ |Q \cap M| = i\}.$$

The transport identity is simply

$$\sum_{c \in Q} (A - tM)_c = \sum_{c \in Q} A_c - t\,|Q \cap M|.$$

The mask-only Phase-P replay in `results/johnson_quaterne_layer_p.json` shows that this language separates the main $176 = 144 + 32$ selected-affine terminal-24 signature from the remaining 60 outside-main records, but does not separate the 144 exact-V_4 records from the 32 selected-affine extras. The anchored replay in `results/johnson_anchor_refinement_p2.json` adds the fixed translation $V_4 = \{0123, 1032, 2301, 3210\}$ from the O5 layer as four Johnson anchors; that anchored profile separates the 144, 32, and 60 classes. The direction-level replay in `results/johnson_anchor_direction_p3.json` explains the exact 144-profile: relative to the fixed translation direction, the terminal affine quaternes have the uniform inventory

$$1 \text{ same direction } + \ 5 \text{ line-intersection directions } + \ 3 \text{ complement directions},$$

giving the anchored shadow

$$4(4,0,0,0) + 20(2,2,0,0) + 12(1,1,1,1).$$

The follow-up replays in `results/johnson_followup_p4.json` and `results/o5e_collision_forbidden_shadow_p5.json` sharpen this boundary. For the exact 144, the 36 terminal-affine quaternes have inner intersection distribution

$$|Q \cap Q'| = 0 : 246, \qquad |Q \cap Q'| = 1 : 192, \qquad |Q \cap Q'| = 2 : 192,$$

and every cell has degree 9. More importantly, the 32 selected-affine extras are exactly the records with positive forbidden quotient-shadow mass. If F_{2110} counts terminal-affine quaternes whose anchored shadow is $(2, 1, 1, 0)$, then

$$F_{2110} = 0 \quad \text{for the exact 144}, \qquad F_{2110} \in \{12, 16\} \quad \text{for the 32 extras.}$$

The linear reason is short. Let $W_0 \leq F_2^4$ be the two-dimensional direction of the fixed translation spread, and let $A = a + L$ be a domain-affine plane. Under the quotient $F_2^4 \to F_2^4/W_0$, the image of A is an affine subspace of dimension 0, 1, or 2; the nonempty fibers of the restricted quotient map all have equal size. Thus the sorted spread-shadow of A is only

$$(4,0,0,0), \qquad (2,2,0,0), \qquad \text{or} \qquad (1,1,1,1),$$

and $(2, 1, 1, 0)$ is impossible for a domain-affine plane. The P6 replay `results/forbidden_shadow_split_p6.json` also shows that the within-the-32 split is not independent: $F_{2110} = 12$ is exactly the 16 two-diagonal translation-subset terminal records, while $F_{2110} = 16$ is exactly the two 8-record V_4-like terminal-set classes. Thus the 144/32 boundary is visible in O5-anchored Johnson data, not in the mask-only Johnson strata.

The 50 transported quaternes are the central bridge between the visible Dürer quaterne layer and the terminal 24-layer. Section 8 shows that 36 of these 50 transported quaternes are affine planes in the F_2^4 tesseract model.

7 Permutation Diagonals and the Type-A Scope

The 24 permutation diagonals of a 4×4 array are indexed by S_4. This is also the chamber set of the type-A Coxeter complex A_3. In the present paper the type-A language is used only at this finite chamber-indexing level.

Certified Proposition 7.1 (Permutation-diagonal subgroup drop)**.** *In the source Dürer-complement square, the target permutation diagonals of sum* 34 *are*

$$\{0123, 0213, 1032, 1302, 2031, 2301, 3120, 3210\},$$

an order-8 subgroup isomorphic to D_4. For every $t > 0$ along the Sagrada ray, the target diagonals of the current magic sum $34 - t$ are

$$\{0123, 1032, 2301, 3210\},$$

the Klein four subgroup V_4.

Certification. This is Claim IDs M24-C10 and M24-C11 in the crosswalk, replayed in `results/magic24_certificate_pack.json`. □

Ray position	Target diagonal set	Group type
$t = 0$	$0123, 0213, 1032, 1302, 2031, 2301, 3120, 3210$	D_4
$t > 0$	$0123, 1032, 2301, 3210$	V_4

Table 4: The Dürer/Sagrada permutation-diagonal drop.

Certified Proposition 7.2 (Type-A guardrail)**.** *In the A_3/S_4 chamber model, these D_4 and V_4 subsets are subgroup/coset chamber tilers, not standard poset cones $L(P)$. Their common-halfspace closure is all of S_4, and their induced chamber graphs are disconnected.*

Certification. This is Claim IDs M24-C12, M24-C41–M24-C43, and M24-C49, replayed in

- `docs/TYPE_A_SUBGROUP_TILERS.md`
- `docs/TYPE_A_POSET_CONE_COMPARISON.md`
- `results/s4_chamber_fingerprints.json`
- `results/s4_poset_cone_comparison.json`

□

Remark 7.3. The general subgroup/coset chamber-tiler component theorem belongs to the separate Type-A/Tetra paper line. In this paper $D_4 \to V_4$ is used only as a Dürer-side enrichment and as a finite guardrail: subgroup tilers are not automatically poset cones.

8 The Binary Tesseract Layer

Sudbery's tesseract reading of Dürer's square suggests labelling the 16 cells by $D_{ij} - 1$, hence by the elements of $\mathbb{F}_2^4$ [Sud13]. In the orientation used here this is not merely a set-theoretic labelling. It is linear in row and column bits.

Certified Proposition 8.1 (Linear tesseract coordinatization)**.** *Write the row index and column index as two-bit vectors*

$$r = (r_0, r_1), \qquad c = (c_0, c_1).$$

If

$$D_{r,c} - 1 = (\ell_0, \ell_1, \ell_2, \ell_3) \in \mathbb{F}_2^4,$$

then

$$\ell_0 = r_1 + c_0 + c_1, \quad \ell_1 = r_0 + c_0 + c_1, \quad \ell_2 = r_0 + r_1 + c_0, \quad \ell_3 = r_0 + r_1 + c_1,$$

with all operations in $\mathbb{F}_2$. Thus the cell-value map is a rank-4 linear automorphism of $\mathbb{F}_2^2 \oplus \mathbb{F}_2^2$. Consequently all 24 permutation diagonals are affine planes in the Dürer F_2^4 model.

Certification. This is Claim IDs M24-C73 and M24-C74, replayed in

- `docs/F2_TESSERACT_LAYER.md`
- `results/f2_tesseract_analysis.json`
- `tests/test_f2_tesseract.py`

The final assertion also uses $\mathrm{AGL}(2,2) \cong S_4$: every permutation of the four column labels is affine over $\mathbb{F}_2^2$, so each graph $x \mapsto \sigma(x)$ is an affine plane. ☐

Certified Proposition 8.2 (Affine source and terminal transport)**.** *There are* $140 = 35 \cdot 4$ *affine planes in* $\mathbb{F}_2^4$*. Inside* $H_{34}(D)$*, the affine quaternes are exactly*

$$13 \textit{ balanced directions} \times 4 \textit{ cosets} = 52.$$

The Sagrada mask has direction W*. Among the* 13 *balanced directions,* 9 *are complementary to* W*, and these contribute*

$$9 \cdot 4 = 36$$

terminal affine quaternes at $D(10)$*. These* 36 *are precisely the pure transport layer from source sum* 34 *and Sagrada incidence* 1*.*

Certification. This is Claim IDs M24-C65–M24-C67 and M24-C75–M24-C77. The replay is in `docs/F2_TESSERACT_LAYER.md` and `results/f2_tesseract_analysis.json`. A direction is balanced exactly when no coordinate bit is constant on it; these are the 13 directions whose cosets have integer value sum 34. Complementarity to the Sagrada direction gives one Sagrada point in each coset, hence terminal sum 24 after subtracting 10. ☐

F_2^4 layer	Count
Ambient affine planes	$35 \cdot 4 = 140$
Balanced source directions	13
Affine source quaternes in $H_{34}(D)$	$13 \cdot 4 = 52$
Balanced directions complementary to Sagrada	9
Terminal affine quaternes in $H_{24}(D(10))$	$9 \cdot 4 = 36$

Table 5: Finite-affine summary of the tesseract transport layer.

The same F_2^2 language gives a concise interpretation of Proposition 7.1:

$$D_4 = \{x \mapsto x + b\} \cup \{x \mapsto \mathrm{swap}(x) + b\}, \qquad V_4 = \{x \mapsto x + b\}.$$

The ray collapses the target diagonal family from translations plus bit-swap translations to translations alone.

Remark 8.3. The F_2^4 layer enriches the Dürer/Sagrada side of the meet. It does not make 24 universal, and in the full order-four atlas it tracks the exact canonical V_4 affine subclass rather than only the historical Dürer/Sagrada cell.

9 Bounded-Polytope Guardrail

Magic squares can also be studied as lattice points in polyhedral cones and bounded polytopes [War80, Ahm04, ADLH02]. Here that language serves mainly as a guardrail.

Certified Proposition 9.1 (The terminal square is not a vertex)**.** *The terminal square $D(10)$ lies on the boundary of the bounded order-four magic-square polytope, but it is not a vertex of either the free-sum bounded polytope or the fixed-sum $S = 24$ bounded polytope. The full fixed-sum $S = 24$ bounded polytope has affine dimension 7 and 292 vertices, with 180 integral and 112 semi-integral vertices. The point $D(10)$ has a two-vertex barycentric certificate*

$$D(10) = \frac{4}{5}V_{126} + \frac{1}{5}V_{140}$$

in the full vertex catalogue.

Certification. This is Claim IDs M24-C52, M24-C56, M24-C60, M24-C61, and M24-C63, replayed in the bounded-polytope and fixed-sum audit artifacts:

- `docs/BOUNDED_MAGIC_POLYTOPE.md`
- `docs/FIXED_SUM24_POLYTOPE_AUDIT.md`
- `results/bounded_magic_polytope.json`
- `results/fixed_sum24_polytope_audit.json`

□

Thus the 4×4 side is a bounded ray endpoint, not a shared vertex mechanism with the Lo Shu side. This negative statement prevents the paper from replacing one overclaim by another.

10 Order-Four Extension

The Dürer/Sagrada endpoint is not unique in the wider normal order-four dataset. The companion replay first reconstructs the classical counts

$$7040$$

raw normal order-four magic squares and

$$880$$

essential representatives [OB82].

Certified Proposition 10.1 (Endpoint-24 atlas and exact V_4 subclass)**.** *Across the* 880 *essential order-four representatives and the eight admissible one-incidence masks, endpoint* 24 *occurs in* 236 *essential square-mask pairs. In those terminal-24 records, affine cell-value labelling occurs in exactly* 144 *pairs, and this equals the exact canonical V_4 terminal subclass in the chosen essential orientation. The main inside-out terminal signature is a broader* 176*-pair class:*

$$176 = 144 + 32.$$

Certification. This is Claim IDs M24-C15–M24-C17, M24-C78–M24-C80, M24-C124–M24-C127, and M24-C139. The replay artifacts are

- `data/order4_normal_essential_880.json`
- `results/order4_endpoint_spectrum.json`
- `results/order4_f2_extension.json`
- `results/exact_v4_affine_class_audit.json`

- `results/inside_out_main_signature_split.json`

□

The conclusion is qualitative rather than uniqueness-based. Dürer/Sagrada is a historical canonical representative inside a larger structured terminal-24 landscape. The 32 extras in the $176 = 144 + 32$ signature are structured near misses: earlier finite invariants did not explain them, while the current O5-anchored Johnson refinement separates them from the exact V_4 class by positive forbidden quotient-shadow mass. Their broader conceptual role remains a follow-up question rather than a strong-meet mechanism.

236	=	$176 \sqcup 60$	terminal-24 records split into main and outside-main
176	=	$144 \sqcup 32$	main class split into exact V_4 and extras
144	⊃	8	the historical Dürer/Sagrada quotient cell.

Figure 3: Order-four terminal-24 containment diagram. The most stable finite class detected by the atlas is the 144-pair globally affine exact-V_4 subclass; the historical Dürer/Sagrada cell is a small canonical representative inside it.

11 Certificate Architecture

The main body uses a narrow claim set. Table 6 summarizes the claim levels and keeps the computational layers in their proper roles.

Layer	Paper role	Primary repository control
Strong meet	theorem	`CLAIM_CROSSWALK.md`
Lo Shu and Sagrada ray	certified propositions	certificate pack
Quaterne transport	certified proposition	certificate pack, inside-out audit
$D_4 \to V_4$, Type-A scope	certified propositions	S_4 reports
F_2^4, bounded polytope	certified propositions / guardrails	Phase F/E reports
880/236/144/176 extension	certified proposition / appendix	Phase C/H/O reports
APD, Markov, Hilbert, N-poset	appendix audits	reproducibility package

Table 6: Paper-v2 claim levels and repository controls.

The strongest methodological lesson is that finite certificates are most useful when they also state what they do *not* prove. This is why the APD/PTE, Morse-Hedlund, Markov, Hilbert, and pointed five-state analyses remain appendices or follow-up problems.

A Endpoint Spectrum for the 880-Representative Atlas

The endpoint spectrum across the 880 essential representatives and the eight one-incidence masks is

endpoint	22	23	24	25	26	27	28	29	30	31	32	33	34
count	176	44	236	268	676	108	328	376	772	420	884	992	1760

Endpoint 24 is therefore not rare in the strongest possible sense. Its role in this paper comes from the strong meet and from the Dürer-side structure, not from absolute uniqueness in the order-four atlas.

The enriched filter ladder

$$7040 \to 236 \to 220 \to 160 \to 152 \to 144$$

passes through endpoint 24, subgroup terminal diagonals, Klein-four profile, terminal quaterne count 96, selected values $\{11, 12, 15, 16\}$, and exact canonical V_4.

B The Exact Canonical V_4 Subclass and the Dürer/Sagrada Cell

The 144-pair exact canonical V_4 subclass has selected values

$$(11, 12, 15, 16),$$

terminal diagonal set

$$\{0123, 1032, 2301, 3210\},$$

and terminal quaterne decomposition $25 + 50 + 21$.

The Phase-O structural audit sharpens this class inside the 236 terminal-24 records. The following conditions are equivalent there:

$$\begin{gathered}\text{exact canonical } V_4,\\ \text{global affine cell-value labelling},\\ \text{zero affine defect},\\ \text{full translation terminal } V_4,\\ \text{preservation of all 140 domain affine planes},\\ \text{preservation of all 24 permutation-diagonal affine planes.}\end{gathered}$$

The selected-mask affine condition alone remains too broad: it gives the $176 = 144 + 32$ main class. The 32 selected-affine extras are the first near-miss layer, with affine defect 4, 76/140 domain affine planes preserved, and 8/24 permutation-diagonal affine planes preserved.

There is also a wider affine-layer explanation of the number 144. It is useful to state it separately, because it no longer depends on filtering the terminal-24 atlas.

Lemma B.1 (Balanced directions)**.** *In $\mathbb{F}_2^4$, exactly 13 two-dimensional linear subspaces are balanced, meaning that no coordinate bit is constant on the subspace.*

Proof. There are

$$\begin{bmatrix}4\\2\end{bmatrix}_2 = 35$$

two-dimensional subspaces of $\mathbb{F}_2^4$. A direction is bad exactly when it is contained in at least one coordinate hyperplane. Each coordinate hyperplane is $\mathbb{F}_2^3$ and contains

$$\begin{bmatrix}3\\2\end{bmatrix}_2 = 7$$

two-dimensional subspaces. Two coordinate hyperplanes intersect in one two-dimensional coordinate plane, while three coordinate hyperplanes intersect in a line and contain no two-dimensional subspace. Hence

$$\#\{\text{bad directions}\} = 4 \cdot 7 - \binom{4}{2} = 22,$$

so $35 - 22 = 13$ directions are balanced. $\square$

Certified Proposition B.2 (Affine normal count)**.** *There are* 432 *essential normal order-four magic squares whose cell-value labelling* $x \mapsto Q_x - 1$ *is globally affine over* $\mathbb{F}_2^4$*. Moreover, across the eight admissible one-incidence masks, these representatives give*

$$432 \cdot 8 = 3456$$

affine square-mask pairs, uniformly distributed as

$$24 \textit{ selected value planes} \times 144 \textit{ pairs.}$$

Certification and finite-linear derivation. Write a globally affine normal square as

$$Q_x - 1 = Ax + b, \qquad A \in GL(4, 2), \quad b \in \mathbb{F}_2^4.$$

If a four-cell affine plane has balanced direction, then every coordinate bit is 1 on exactly two of its four labels. Therefore its label sum is

$$2(8 + 4 + 2 + 1) = 30,$$

and its value sum is $30 + 4 = 34$. Thus a linear part A is good exactly when it sends the row, column, and diagonal directions of the cell domain to balanced directions in value space; the main and anti-diagonal are parallel affine planes and share the same direction.

The finite F_2-linear replay gives the following count. The complementarity graph on the 13 balanced directions has 6 triangles, so there are $6 \cdot 6 = 36$ ordered pairwise-complementary balanced image triples. For each ordered image triple there are 6 linear maps, hence

$$36 \cdot 6 = 216$$

good linear parts. Offsets are free, giving

$$216 \cdot 16 = 3456$$

raw affine normal squares. Since a normal square has all entries distinct, no nontrivial square symmetry fixes it cellwise; hence each square-symmetry orbit has size 8, and

$$3456/8 = 432$$

essential affine representatives.

The same replay shows that each affine representative sends the eight admissible masks to exactly

$$8 = 2 \text{ coordinate directions} \times 4 \text{ cosets}$$

selected value planes. The two directions form one of the three matchings of the four coordinate axes:

$$1, 2 \,|\, 4, 8, \qquad 1, 4 \,|\, 2, 8, \qquad 1, 8 \,|\, 2, 4.$$

The 24 value-bit permutations preserve the affine normal layer and induce the natural $S_4 \to S_3$ action on these three matchings. Each matching has stabilizer size 8, so the 432 affine representatives split as

$$144 + 144 + 144.$$

This yields the uniform 24×144 selected-value-plane split.

The replay artifacts are

- `results/affine_normal_layer.json`
- `results/affine_normal_spread_structure.json`

- `results/affine_normal_count_derivation.json`

$\square$

Lemma B.3 (Admissible-mask torsor)**.** *The square-symmetry group of order* 8 *acts freely and transitively on the eight admissible one-incidence masks.*

Certification. The eight masks are

$$0231,\ 0312,\ 1203,\ 1320,\ 2013,\ 2130,\ 3021,\ 3102.$$

The square-symmetry replay has one orbit of size 8 and trivial stabilizer for each mask. This is recorded in `results/affine_normal_count_derivation.json`. $\square$

The two appearances of 3456 count different but compatible sets:

$$\begin{array}{rcll} 216 \cdot 16 & = & 3456 & \text{raw affine normal squares,} \\ 432 \cdot 8 & = & 3456 & \text{essential affine square-mask pairs.} \end{array}$$

Lemma B.3 explains why these counts mirror each other after passing between oriented squares and essential representatives with a chosen admissible mask.

Finally, if a selected value plane is P, then its bounded downward endpoint is

$$34 - (\min P - 1) = 35 - \min P.$$

Among the 24 selected value planes in Proposition B.2, the unique plane with endpoint 24 is

$$\{11, 12, 15, 16\}.$$

Every record in this plane has full translation terminal V_4. Thus the '144' is not only a terminal-24 census count: it is one uniform value plane inside the globally affine normal order-four layer.

Certified Proposition B.4 (Affine translation torsors)**.** *For every one of the* 3456 *globally affine square-mask pairs, the four fixed translation diagonals*

$$0123, \qquad 1032, \qquad 2301, \qquad 3210$$

map to the four cosets of one balanced value-space direction W. If $P = a + U$ is the selected value plane, then

$$\mathbb{F}_2^4 = U \oplus W,$$

so P intersects each W-coset in exactly one point. The resulting map from the four translation labels to the four points of P is an affine bijection.

Moreover, this structure is uniform over all selected value planes. For each selected direction U, let Q be its coordinate partner in the matching of coordinate axes. The nine balanced complements to U are graphs of linear maps

$$Q \longrightarrow U$$

with no zero rows. They split as

$$9 = 6 + 3,$$

where the six used directions are precisely the invertible maps $\mathrm{GL}(2,2)$*, and the three unused directions are rank-one graphs. Hence each selected plane has*

$$6 \text{ used directions} \times 24 \text{ records} = 144$$

affine square-mask pairs.

Certification. This is a finite replay of the Phase-O5 torsor refinement. The kernel guardrail is recorded in `results/kernel_v4_o5.json`: the value-bit kernel V_4 of the $S_4 \to S_3$ action on coordinate matchings is not the terminal translation V_4. The quotient-section statement is replayed in `results/terminal_parallel_torsor_o5b.json`, where all 3456 affine square-mask pairs have balanced translation-image direction, complementary selected direction, intersection pattern $(1, 1, 1, 1)$, and affine translation-to-selected point map.

The six-of-nine split for the terminal-24 plane is replayed in `results/six_of_nine_o5c.json`, and the all-selected-plane generalization is replayed in `results/selected_plane_six_of_nine_o5d.json`. There are 24 selected planes; each has 144 records, nine balanced complements to its selected direction, six invertible graph complements, three rank-one graph complements, and zero violations of the rule that the used directions are exactly the six invertible graph complements. □

Remark B.5 (The final 24-factor)*.* The count

$$3456 = 24 \text{ selected planes} \times 6 \text{ invertible directions} \times 24 \text{ records}$$

does not mean that every (P, W)-fiber is the full $\mathrm{AGL}(2, 2)$-torsor of affine point maps. The Phase-O5e/O5f audit shows that only 96 of the 144 (P, W)-fibers realize all 24 affine point maps; 48 fibers have duplicated point maps. The safe parametrization is mask-refined: $(P, W, \phi, \text{mask})$ is injective, with 24 distinct (ϕ, mask) pairs in every (P, W)-fiber. The P5 replay sharpens the negative statement: in every (P, W)-fiber the point-map collision graph is a matching with 0, 4, or 8 collision edges, and the pair consisting of selected-plane direction and translation direction determines which case occurs. The replays are `results/torsor_parametrization_o5e_o5f.json` and `results/o5e_collision_forbidden_shadow_p5.json`.

Source type	Count	Source diagonal set
S1	96	$\{0123, 1032, 2301, 3210\}$
S2	16	$\{0123, 0132, 1023, 1032, 2301, 2310, 3201, 3210\}$
S3	16	$\{0123, 0213, 1032, 1302, 2031, 2301, 3120, 3210\}$
S4	16	$\{0123, 0321, 1032, 1230, 2103, 2301, 3012, 3210\}$

Table 7: Source diagonal types inside the 144-pair exact canonical V_4 subclass. The Dürer/Sagrada representative lies in type S3.

The Dürer/Sagrada quotient cell has 8 pairs. In the current finite atlas it is characterized by APD vector $(0, 0, 0, 55296)$, source type S3, exact terminal V_4, and terminal quaterne decomposition $25 + 50 + 21$. The essential square indices are

$$112,\ 174,\ 289,\ 359,\ 789,\ 802,\ 834,\ 849.$$

All 8 records are associated and complement-fixed after canonicalization. The canonical Dürer-complement representative is square index 174; under canonicalization the Sagrada mask appears as 1203, with selected values $12, 11, 15, 16$.

C Inside-Out, APD, and Permutation-Polytope Audits

The following audits support the paper while staying outside the main theorem mechanism.

- **Inside-out set systems.** The row, column, diagonal, mask, source-quaterne, and terminal-quaterne incidence systems have exact rank/SNF/kernel and colored-automorphism replays. The main terminal signature is the broad $176 = 144 + 32$ class; the later O5-anchored Johnson replay separates its 144 exact-V_4 records from the 32 extras by the forbidden quotient-shadow count F_{2110}.

- **APD/PTE.** Along the Sagrada ray,

$$\mathrm{APD}_1(t) = 0, \qquad \mathrm{APD}_2(t) = 0, \qquad \mathrm{APD}_3(t) = -24t(t-4)(t-16).$$

  Moreover $\mathrm{APD}_3(D(t)) = 6 \det \widehat{E}(t)$ in the centered local model. This gives a useful parity-power or Prouhet–Tarry–Escott reading of the ray [Tak25, Tak26], but endpoint 24 is selected by bounded terminality, not by APD vanishing.

- **Morse-Hedlund/Prouhet family fit.** The Dürer-complement source fits the displayed Morse-Hedlund/Prouhet order-four family, but the Sagrada direction, terminal point, and full ray do not under the tested transform scope [Ser11].

- **Permutation polytopes.** The subgroup polytopes

$$P(D_4) = \mathrm{conv}\{P_\sigma : \sigma \in D_4\}, \qquad P(V_4) = \mathrm{conv}\{P_\sigma : \sigma \in V_4\}$$

  fit naturally into the permutation-polytope language [BHNP09, GP06]. The face audit shows that $P(V_4)$ is a tetrahedron, but it is not a face of $P(D_4)$ or of the Birkhoff polytope B_4.

D Markov and Hilbert Follow-Up Audits

The terminal-24 records also support two technical follow-up directions.

- The first Markov-style graph uses primitive $\{-1, 0, 1\}$ fixed-sum diagonal-magic kernel moves. It has 236 nodes, 247 edges, 98 connected components, and 50 isolated vertices. This is a small-move graph, not a complete Markov-basis theorem.

- The finite Hilbert-style audit through magic sum 8 finds 20 checked atoms:

$$8 \text{ at sum } 1, \qquad 12 \text{ at sum } 2.$$

  All 236 terminal-24 endpoints decompose in this checked atom set. This is not yet a proof of the full Hilbert basis of the nonnegative diagonal-magic cone.

E The Pointed Five-State Bridge

The Lo Shu $S = 24$ fiber has five points. A separate audit compares this five-state object with the graph of linear extensions of an N-poset and with the order-ideal lattice $J(A_2^+)$, using standard poset language [Naa00, Sta86].

The natural Lo Shu adjacency is a diamond with a central point and four boundary points, so it is not directly the N-poset linear-extension graph. After adding a boundary cycle on the four nonconstant points and marking one boundary point, however, the resulting pointed graph is a 4-cycle with one pendant vertex. It is isomorphic to the N-poset linear-extension graph and to the Hasse graph of $J(A_2^+)$.

The bridge is controlled but noncanonical: every boundary mark works, and no current Dürer-side structure selects one. It remains an open problem rather than a foundation for the meet theorem.

F Open Problems

The most useful follow-up questions are:

1. Extend the finite-linear Phase-O/O2/O3/O4/O5 explanation of the exact V_4 affine class beyond the normal order-four affine layer, or show that this layer is the natural boundary of the phenomenon.

2. Generalize the forbidden quotient-shadow obstruction for the 32 structured extras inside the $176 = 144 + 32$ inside-out signature, or show that it is specific to this fixed order-four selected-affine terminal layer.
3. Develop subgroup/coset chamber tilers as a sibling theory to type-A poset-cone tilers. This belongs primarily to the Type-A/Tetra paper line; Magic supplies the $D_4 \to V_4$ seed example.
4. Decide whether a complete Markov-basis or Hilbert-basis treatment is worth a separate technical paper.
5. Find, or rule out, a natural Dürer-to-Lo-Shu boundary selector for the pointed five-state bridge.

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